Hybrid Tensor-Train Time Evolution: Status, Findings, and Outlook

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TN_Turbulence project — 2D passive scalar study
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Concise status note on the machinery built to evolve a 2D scalar field in the hybrid tensor-train (TT) encoding of the compression study. Records the equations being solved, what has been implemented, what has been checked, the findings and caveats so far, and the plan going forward. Detailed numbers live in the test suite; this note keeps only the conclusions.

I. THE EQUATIONS

A. Passive scalar advection–diffusion

The target problem, and the one the compression study characterises statistically:

$$\partial_t\theta + \mathbf{u}\cdot\nabla\theta \;=\; \kappa\nabla^2\theta + f, \qquad \nabla\cdot\mathbf{u} = 0, \tag{1}$$

on a doubly periodic square. Here θ is a scalar (temperature, dye concentration) carried passively by a prescribed incompressible velocity $\mathbf{u}$: it does not react back on the flow. κ is the molecular diffusivity and f a large-scale source.

a. Physics. Advection stretches and folds θ, cascading scalar variance from the injection scale down to the Batchelor scale where diffusion erases it. The resulting fields are strongly *intermittent*: increments have heavy tails and flatness grows without bound as the separation shrinks, which is precisely the property the hybrid encoding was designed to preserve. The variance budget follows from Eq. (1),

$$\frac{d}{dt}\,\tfrac{1}{2}\langle\theta^2\rangle \;=\; -\,\kappa\big\langle|\nabla\theta|^2\big\rangle \;+\; \langle\theta f\rangle, \tag{2}$$

because the advective term is skew-symmetric for $\nabla\cdot\mathbf{u} = 0$ and therefore conserves $\langle\theta^2\rangle$ exactly. Note the consequence, which matters for scheme design: the *nonlinear* (bilinear) transport term is the conservative one, and the *linear* diffusive term is what dissipates. The same structure holds for Navier–Stokes, where $\mathbf{u}\cdot\nabla\mathbf{u}$ conserves kinetic energy and viscosity dissipates it.

When $\nabla\cdot\mathbf{u} = 0$ the transport term also admits the *conservative* (flux) form

$$\mathbf{u}\cdot\nabla\theta \;=\; \nabla\cdot(\mathbf{u}\,\theta), \tag{3}$$

which is equivalent analytically but not after discretisation: the flux form respects the variance budget far better and is the usual choice.

B. Cases solved so far

Work to date has deliberately used constant-coefficient reductions of Eq. (1), because they have exactly known solutions and therefore isolate compression error from every other error source:

$$\partial_t\theta = \kappa\nabla^2\theta \qquad \text{pure diffusion} \tag{4}$$

$$\partial_t\theta + \mathbf{c}\cdot\nabla\theta = 0 \qquad \text{rigid translation, } \mathbf{c} \text{ constant} \tag{5}$$

$$\partial_t\theta + \mathbf{c}\cdot\nabla\theta = \kappa\nabla^2\theta \qquad \text{constant-velocity advection–diffusion} \tag{6}$$

The next step introduces a genuine, space-dependent velocity — the steady cellular flow $\mathbf{u} = (\sin y,\ \sin x)$ — which is incompressible, analytically known, and generates fine scales exponentially.

C. Standard ways to solve these

- **Pseudo-spectral** is the reference method for periodic 2D turbulence: derivatives are exact in Fourier space, the nonlinear product is formed in physical space with 2/3 dealiasing, and diffusion is handled by an integrating factor or exponential time differencing so it imposes no timestep restriction. This is how the DNS database underlying the paper was produced.

- **Finite difference / finite volume** with upwind or WENO reconstruction for the transport term, central stencils for diffusion, and SSP Runge–Kutta in time. Cheaper per degree of freedom, more dissipative, and much easier to express as a local operator.
- **Stiffness.** Explicit diffusion requires $\Delta t \leq \Delta x^2/4\kappa$ in 2D, which becomes punishing on fine grids; implicit or exponential treatment of the diffusive term is standard.

a. What we do instead. The field is stored as a hybrid TT over the interleaved (Morton) bit ordering, and every spatial operator is a matrix-product operator acting on it. Stencils are finite differences, so the semi-discrete system is

$$\frac{d\boldsymbol{\theta}}{dt} \;=\; \mathsf{A}\,\boldsymbol{\theta}, \qquad \mathsf{A} \;=\; -c_x \mathsf{D}_x - c_y \mathsf{D}_y + \kappa\,\mathsf{L}, \tag{7}$$

with A collapsing to a *single* low-bond MPO in the constant-coefficient case. Time stepping is explicit Runge–Kutta, and the state is recompressed after every stage. Compression is the entire source of novelty and the entire source of risk.

II. THE ENCODING

Writing the hybrid ansatz out,

$$\theta(x,y) \;=\; \sum_{\alpha=1}^{\chi} A^0[B,\alpha]\;\Phi_\alpha(\text{fine coordinates}), \tag{8}$$

with B the coarse-block label carried by the dense head and Φ_α the fine-scale shape functions carried by the compressed tail. So the hybrid TT is a **block-wise reduced basis with a globally shared dictionary**. Two consequences:

- The coarse-grained field is a linear functional of the head alone, so "the large scales live in the head" is literally true.
- The large scales carry no rank error. For advection-dominated dynamics this is the central argument for the encoding: exact large-scale transport with rank-compressed small-scale structure.
- The same framing predicts the *seam*: neighbouring blocks get independent amplitude vectors and nothing enforces continuity across a block boundary.

a. Conventions. L = bits per axis (grid $2^L \times 2^L$); $2L$ total interleaved legs, even legs carrying rows (y) and odd legs columns (x); N = legs held by the dense head, so the head spans $2^{N/2} \times 2^{N/2}$ coarse blocks and the seam period is $2^{L-N/2}$ — *not* $2^{N/2}$, which coincides only at $N = L$.

III. WHAT WAS BUILT

All of it sits on top of `tensnet` via its public API. No `tensnet` source is modified, and — deliberately stricter — no `tensnet` global state is read or written, since a wrong global would not raise but would silently produce wrong physics.

- `hybrid_ops.py` — hybrid TT algebra: construction and contraction, truncated SVD with a genuine amplitude tolerance, two-sweep compression with *per-bond* caps, exact arithmetic (add / scale / axpy / Hadamard), MPO application by un-fusing the head, application through an explicit fused $2^N \times 2^N$ head operator, rank-1 mass projection, and pointwise evaluation without contracting the TT.
- `hybrid_operators.py` — named stencils, derivative and Laplacian MPOs, exact translation, the combined advection–diffusion operator, exact MPO algebra, and the isotropic seam smoother.
- `hybrid_evolve.py` — Euler / RK2 / SSPRK3 / RK4, a truncation policy that records the variance it removes, a stable-timestep helper enforcing both the advective and diffusive limits, dense finite-difference and analytic spectral references, and diagnostics.
- Four test suites, ∼117 assertions, all against dense references.

The seam smoother is the isotropic repair — the `_int64_xy` treatment, not the x-only one. With $P_{x,y}$ the seam projectors, $Q = I - P$, and $A_{x,y}$ the periodic two-neighbour averages,

$$S \;=\; Q_x Q_y \;+\; P_x Q_y A_x \;+\; Q_x P_y A_y \;+\; P_x P_y \tfrac{1}{2}(A_x + A_y). \tag{9}$$

IV. WHAT HAS BEEN CHECKED

Everything below is verified numerically against a dense reference, not asserted from reading source.

- Lossless hybrid round trip; agreement between our constructor and the legacy one; the index layout and seam geometry.
- All arithmetic is exact, with no hidden truncation.
- Compression: lossless at full rank, monotone in the cap, head physical dimension preserved, head bond capped independently of the tail bonds, and norm genuinely lost when truncating.
- Every operator — first and second derivatives at 2nd and 4th order, upwind stencils, Laplacian, translation, seam smoother — against dense, on both axes, at several (L, N).
- Operator composition, and agreement between composing then applying versus applying twice.
- The seam smoother reproduces the dense reference implementation exactly, is isotropic under transposition, and touches only seam lines.
- All four time steppers reproduce their dense counterparts.
- The fused-head and un-fused-head paths agree to round-off, both as single applications and over multi-step evolutions.
- Pointwise evaluation agrees with the dense field.

a. Verified `tensnet` traps. None of these raise; all silently corrupt. The global 2D architecture defaults to the non-interleaved layout; `get_diff_mpo` defaults to 8th-order stencils; MPS addition and multiplication silently compress at a global l_2 tolerance; `tt.svd` treats its `eps` as a squared-energy fraction, so the achievable amplitude error saturates around $\sqrt{\epsilon}$; `tt.svd` renormalises singular values after truncation by default; `shift_mpo` has inverted flags and is non-periodic; `MPO.__matmul__` and `MPO.contract()` cannot handle hybrid shapes at all.

V. FINDINGS AND REMARKS

- **The low-rank structure survives advection.** A one-cell translation inflates the state rank by of order ten percent, transiently, and returns it *exactly* to baseline whenever the shift is a whole number of coarse blocks. Verified at two different (L, N). An adaptive-tolerance scheme tracks the intrinsic rank of the exact solution to within one or two, so the method accumulates essentially no spurious rank.
- **Truncation error saturates rather than compounding.** Under a hard rank cap over hundreds of advection steps the error stops growing and the field settles onto a representable state that then simply translates. This was the principal risk identified at the outset and it did not materialise for constant-velocity transport.
- **Lossy truncation per step is a property of the cap, not of advection.** It appears when the rank cap sits below the field's intrinsic rank — which is exactly the fixed-compression-ratio regime the study works in, and a legitimate choice, but the loss should be attributed correctly.
- **Rank capping and tolerance capping answer different questions.** A cap fixes the compression ratio and lets error float; a tolerance fixes error and lets rank float, revealing what the physics actually demands. Both are supported and both are needed.
- **The seam smoother should not run inside the time loop.** It is a low-pass filter and is not idempotent, so iterating it is diffusion. It never beat leaving it off. The *severity* is configuration dependent — it scales with the fraction of cells on seam lines, $\approx 2^{1-L+N/2}$ — so the earlier dramatic figures came from an unrepresentatively small block width. The seam is an artifact of the representation, so its repair belongs on the output being analysed, not on the state being evolved.
- **Mass conservation is cheap and worth having.** Truncation drifts the mean; a rank-1 constant restores it to round-off at no accuracy cost, provided it is applied *after* truncation. Unlike rescaling the l_2 norm it does not move variance between scales.
- **Do not renormalise after truncation.** Truncation is an orthogonal projection and losing norm is a theorem, not a defect. Rescaling puts the discarded small-scale variance back into the retained large scales — a spurious inverse cascade — and, worse, disables the very diagnostic that detects under-resolution.
- **Consequence for the published data.** Because rescaling does not change the retained subspaces, the legacy pipeline's output is exactly plain truncation times a *global scalar* that tends to unity as the bond dimension grows. Every scale-invariant statistic — flatness, normalised structure-function ratios, normalised increment PDFs — is therefore unchanged; only the absolute level of $E(k)$ can have moved. This still needs confirming on the real DNS snapshots.

- **The fused head does not scale.** Single-stencil operators have very sparse fused heads, but operators that are sums of terms — the Laplacian, the seam smoother — have essentially dense ones, so sparsity buys nothing for exactly the operators one most wants. It is modestly faster at small N but becomes infeasible in memory at production N; the un-fusing path, whose cost carries no 4^N term, is the one that scales.
- **Two traps for the experimenter.** Rank experiments need a field whose rank has room to move — a multiscale field is full rank and the measurement merely saturates the chain geometry. And a total error that falls *below* the pure finite-difference floor is coincidental error cancellation, not an improvement.

VI. CORRECTIONS TO EARLIER STATEMENTS

Recorded because the reasoning matters more than the conclusions:

- "Translation is rank-preserving" (wrong), then "translation is not rank-preserving, so every step must be lossy" (also wrong, over-corrected). The measured behaviour is the first bullet of the previous section.
- "Apply the seam smoother once per step, inside the loop." It is harmful at every cadence tested.
- "The nonlinear term is the norm-changing one." For a passive scalar with $\nabla\cdot\mathbf{u} = 0$ the transport term conserves variance exactly and the linear diffusive term dissipates; see Eq. (2).

VII. OUTLOOK AND NEXT STEPS

1. **Commit the validated baseline.** Nothing is under version control yet.

2. **Cellular flow $\mathbf{u} = (\sin y,\ \sin x)$ — the immediate next step.** Requires the velocity components as hybrid TTs (analytic, low rank), the bilinear $\mathbf{u}\cdot\nabla\theta$ path via the Hadamard product, and both the advective and conservative forms of Eq. (3). *Settles:* whether the rank saturates or grows without bound once there is a genuine cascade — the question the whole programme rests on, and one that constant-velocity transport cannot test. *Risk:* this is where it could fail; the Hadamard product multiplies ranks before every recompression and is also the cost driver.

3. **Numerical versus physical dissipation.** Convert the logged truncation loss into a dissipation rate and compare with $\kappa\langle|\nabla\theta|^2\rangle$ from Eq. (2). Cheap, reuses the previous runs, and decides on evidence whether a norm-conserving integrator is needed at all.

4. **Hybrid versus standard TT at matched compression.** The actual scientific claim and the true sequel to the paper: does the hybrid encoding reproduce intermittency statistics *under evolution* better than a standard TT at equal ρ? Needs a standard-TT evolution path, the existing degree-of-freedom matching, and the paper's diagnostics recast as functions of time. Compression quality does not imply dynamical quality; this is what measures it.

5. **Scale to the production geometry** (4096^2). Use the un-fusing head path. Performance has never been profiled and will need attention.

6. **Real DNS velocity fields**, together with the deferred check of the renormalisation effect on the published figures. Both need the dataset.

7. **Conditional: TDVP or implicit stepping.** If step 3 shows truncation dissipation dominating, the principled remedy is tangent-space integration, which projects the generator rather than the state and so conserves the norm and any quadratic invariant of the generator exactly. Deliberately gated on a measurement rather than adopted speculatively. Implicit or exponential treatment of diffusion becomes necessary if the timestep restriction bites.

a. Beyond the passive scalar. The natural extension is 2D Navier–Stokes in vorticity form, $\partial_t\omega + \mathbf{u}\cdot\nabla\omega = \nu\nabla^2\omega$ with $\mathbf{u}$ recovered from ω by an elliptic solve, which adds a Poisson inversion — a natural fit for the fused/sparse head — and a second quadratic invariant, since 2D conserves both energy and enstrophy and truncation will damage them at different rates.